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The Geometric Origin of the Constant 2 in Ramanujan Prime Asymptotics via Dyadic Generational Decomposition

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Zenodo
2026-05-25 | Conference poster
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CONTRIBUTORS: Priyal Bhagwanani

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Priyal Bhagwanani (Author)

Citation (bibtex)
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Description
This work investigates the asymptotic structure of Ramanujan primes and provides a structural explanation for the appearance of the constant 2 in the classical limit relationRn / (n log n) -> 2.Although this asymptotic behavior follows from existing consequences of the Prime Number Theorem, prior derivations treat the constant primarily as an analytic residue emerging from asymptotic estimation. The present study addresses the unresolved conceptual question of why this specific constant necessarily governs the growth scale of Ramanujan primes.

We re-examine the defining interval condition of Ramanujan primes through a distributional and scaling-based framework that isolates the proportional behavior governing prime generation across expanding intervals. Within this formulation, the factor 2 emerges intrinsically from the structural balance required by the Ramanujan prime condition itself, rather than from secondary cancellation phenomena in asymptotic expansions. Using this perspective, we establish the corresponding asymptotic upper growth behavior for Rn and derive the limiting constant from first-principles structural constraints on admissible prime-counting intervals.

The results provide a conceptual reinterpretation of Ramanujan prime asymptotics by identifying the constant 2 as a geometric-distributional scaling invariant associated with the interval architecture underlying Ramanujan prime generation. This shifts the interpretation of the classical asymptotic law from a purely analytic consequence of prime-counting estimates to a manifestation of deeper structural properties of prime distributions.

Beyond its theoretical significance in analytic and asymptotic number theory, the framework developed here suggests broader relevance for the study of interval-constrained prime phenomena, large-scale prime distribution models, and computational methods that exploit predictable scaling behavior in prime-search and prime-sieving problems.

Added

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Source: Priyal Bhagwanani

The Local Gauge-Persistence Transform for Yang-Mills Gradient Flow: Topological Invariants on Gauge Quotients and Isomorphisms to Protein Folding Dynamics

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Description

We introduce the Local Gauge-Persistence Trans-form (LGPT), a flow-indexed, barcode-valued operator onthe space of Yang-Mills gauge connections. By evaluating thepersistent homology of gradient-flow orbits within symmetry-reduced Sobolev quotients, we extract coordinate-free topologi-cal invariants: total persistence (Ptot), max persistence (Pmax),and persistence entropy (Stop). We reformulate the Yang-Mills mass gap condition via Topological Data Analysis (TDA),establishing a four-way equivalence between the mass gap, H0vacuum persistence, spectral function bounding, and H1 birthvalues in frequency space. We identify a structural dictionarymapping the non-Abelian commutator [Aμ, Aν] to the multi-plicative prime structure represented by the M¨obius function.Furthermore, we establish a mathematical isomorphismbetween Yang-Mills relaxation and the protein folding problem,mapping Sobolev connections to R3N conformational space andgauge symmetry to SO(3) rotational invariants. Experimentalvalidation on synthetic and folding-proxy trajectories admitstopological cooling (entropy dissipation) under gradient flowand simulation computation / preliminary hypothesis testing upon synthetic data achieves 98.00% accuracy in gauge-invariant sector classi-fication. Further experimentation on real datasets / computation / simulation in different environments / prototyping is ongoing. Kindly grant forgiveness for any wrong assumptions to have been made, or room to improve on early mistakes, I am a simple computer science and digtial signal processing engineer striving for opportunities to improve on the work through attempts. I am grateful for suggestions/reviews and honestly look forward to them. Please forgive me for lingual mistakes in positioning mathematical ideas - this is a first write-up whose abstract and introduction were asked to be written by certain LLMs which helped me produce a manuscript before unforeseen deadlines. Thanks for understanding.

Added

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Using Unruh–Bogoliubov Thermal Spectra as a Time-Domain Filter for Inertial Sensors: An Exploratory Study

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Description

This study investigates a nonlinear time-domain filtering approach for inertial sensor drift, utilizing a heuristic bound derived from the Unruh-Bogoliubov thermal spectrum formulation in quantum field theory. The method maps phase increments to a local frequency proxy to evaluate signal-noise discrimination. Preliminary evaluation on synthetic non-Markovian noise demonstrates O(N) computational complexity and performance comparable to standard Extended Kalman Filter (EKF) baselines. Empirical results show comparable estimation error and bias, with reduced computational latency due to a single-pass structure. No claims are made.

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Designing Resilient AI-Powered Navigation Through Domain-Driven Geospatial Persistence

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Description
Conversational AI agents for navigation (e.g., Gemini Live,ChatGPT Voice) are typically implemented as monolithic applicationswhere location tracking and dialogue management share a single pro-cess lifecycle. This architecture creates a critical failure mode: networkdisruptions or AI service crashes cause a complete loss of positionalstate, forcing users to restart from scratch. We present a Domain-Driven Design (DDD) approach that strictly separates the GeospatialTracking Domain (persistent C++ daemon with sensor fusion) fromthe Dialogue Management Domain (transient cloud LLM client). Ourarchitecture uses zero-copy shared memory for inter-domain communi-cation, achieving sub-microsecond context synchronization while main-taining independent lifecycles. In simulated network failure scenarios,our system preserves continuous location tracking through 15-secondoutages and resynchronizes with the LLM client in under 200ms uponreconnection—compared to 5-8 second cold-start delays in monolithicbaselines. This design pattern is broadly applicable to any real-time AIsystem operating on high-frequency sensor streams (robotics, AR/VR,IoT), where persistence and responsiveness must coexist.

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Cryptographic Content-Addressed Version-Aware Distributed Mutual Exclusion

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Description

Distributed mutual exclusion mechanisms range from low-latency centralized systems (e.g., Redis-based locking) to strongly consistent consensus-based services (e.g., etcd and ZooKeeper). While consensus-based approaches provide linearizable guarantees, they incur coordination overhead, quorum latency, and operational complexity, making them inefficient for high-frequency, short-lived locking workloads. In contrast, Redis-based locks achieve low latency but rely on application-defined resource identifiers that are decoupled from execution context, providing no guarantees about the code version or invariants of a critical section. Additionally, systems that compute cryptographic identifiers at runtime introduce non-trivial overhead (10–50 μs per operation), which becomes significant at scale.

We present SC-DKVL, a version-aware distributed mutual exclusion framework that defines lock identity as a deterministic function of code content rather than resource names. By reusing precomputed Git-derived hashes—available directly from the code-hosting layer where every version is already content-addressed—SC-DKVL eliminates runtime hashing and enables constant-time lock identification, while ensuring that distinct code versions map to distinct locks and cannot interfere during rolling deployments.

To operationalize this, a lightweight build-time sidecar script extracts repository-native hashes and generates a header-only C++ mapping that binds code regions to lock identifiers. This establishes a direct bridge between build-time artifacts and runtime synchronization without introducing additional execution overhead. The framework's Redis backend uses atomic Lua scripting and fencing tokens to guarantee correctness under concurrency and failure, achieving 0.5–2 ms lock acquisition latency compared to 3–15 ms for etcd leases.

SC-DKVL avoids consensus for identity management, requiring coordination only for shared state. It supports pluggable backends, including Redis for low-latency execution and consensus systems for durability when needed. By shifting synchronization from resource-level naming to code-level identity, SC-DKVL provides stronger semantic guarantees alongside improved performance, forming a practical foundation for version-aware distributed synchronization in modern microservice systems.

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Description

We present an application-oriented streaming filter built from Ramanujan sums and divisor-mode recursions for real-time periodicity auditing. The method targets observables with cyclotomic or divisor structure and expresses the corresponding kernel through the classical Möbius divisor formula for Ramanujan sums. This yields a fixed-memory online implementation in which each divisor contributes one recursive state. The resulting filter computes an exponentially weighted Ramanujan correlation using a small number of arithmetic states, with per-sample work proportional to the number of divisors of the target order. The contribution of this paper is a streaming architecture based fixed-memory divisor-state recursion system, and lightweight periodicity score suitable for online diagnostics. We illustrate the method on pseudo-random bit streams and discuss its use as a real-time arithmetic periodicity detector. We do not claim superiority over standard statistical tests such as chi-square or sliding spectral methods; rather, we present a distinct arithmetic filtering approach that is simple to implement and useful in practice.

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Analytic Geometry and Mirror Symmetry in Additive Prime Representations

Everyone

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Description

We present the study and results of forming a mathematical framework to reformulate the binary Goldbach conjecture as a measurable phenomenon of symmetric prime geometric offsets. By transitioning from a binary existence framework to a quantitative measurement of symmetry deficit, we define the mirror offset $d(m)$ and gap persistence $\delta(e)$. Computational topology over midpoints up to $m=4000$ reveals highly structured modal behavior, an extreme-bulk suppression dynamic (peak compression), and sublinear growth bounds consistent with $O(\log^2 m)$. To resolve the representation gap of discrete Cartesian metrics, we extend these findings into continuous analytic embeddings. Utilizing Hilbert transforms, we extract smooth attractors from jagged prime data, framing additive prime sequences as continuous analytic signals governed by quasi-periodic phase structures. We also establish a frequency-domain identity linking Goldbach partitions to the power spectrum of the prime indicator sequence, and we present empirical evidence for the non-vanishing of the analytic signal on all tested intervals.

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Generational Emergence of Even Numbers from Cumulative Prime Sumsets: A Structural Analysis Toward the Strong Goldbach Conjecture

Everyone

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Subtitle

Modeling the Dynamic Evolution of Goldbach Representations through Cumulative Filtration

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Priyal Bhagwanani (Author)

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Description

We introduce a cumulative prime-sum filtration to analyze the structured emergence of even integers. By formulating the sumset evolution recursively, we model the representation of even numbers as a \textbf{hitting-time problem} within a growing family of interactions. This approach organizes all even integers into a sequence of \emph{generations} indexed by their first appearance time.

The paper defines these generational classes (Blue, Purple, Brown, and Finalized) precisely and proves foundational structural theorems, including a Trap Property that establishes the permanence of finalized elements and a Frontier-overlap Property via classical prime gap estimates. We provide asymptotic capacity comparisons using the Prime Number Theorem to illustrate the increasing abundance of representations. Furthermore, we catalog why unconditional completeness—the assertion that every even integer up to the finalized boundary appears—reduces to the classical Goldbach conjecture.

We survey partial and conditional approaches, including the circle method and sieve heuristics, and provide a roadmap for rigorous partial results. Finally, we provide appendices detailing investigated analytical approaches, reproduced computational samples, and a reproducible React/TSX implementation to verify these results.

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A Generalized Polyhedral Glyph Framework for Interactive High-Dimensional Visualization

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Logistic Saturation Optimizer: A Reduced-Order Nonlinear Adaptive Optimizer for Bounded Cluster-Wise Gradient Dynamics

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Description

Large-scale optimization in modern machine learning is constrained by the computa-tional and memory costs associated with curvature-sensitive update methods. While first-order adaptive optimizers such as AdamW provide efficient O(N) updates, they can exhibitinstability in highly anisotropic loss landscapes. Approximate second-order methods, suchas Kronecker-factored techniques, address these limitations but introduce substantial com-putational overhead through block-wise matrix inversions.We propose the Logistic Saturation Optimizer (LSO), a bounded nonlinear adaptive op-timizer motivated by reduced-order dynamical systems. LSO models cluster-wise adaptiveupdate dynamics through autonomous logistic saturation flows, constraining update mag-nitudes within a bounded phase-space manifold. This formulation yields a computationallylightweight surrogate for curvature-sensitive adaptation, operating strictly via scalar up-dates.We derive the continuous-time saturation dynamics and establish formal convergenceguarantees to stationary points under standard non-convex smoothness assumptions. Exten-sive empirical evaluations on vision (CIFAR-10/ResNet-18) and language modeling (WikiText-2/Transformer) benchmarks demonstrate that LSO matches or exceeds the convergencequality of

AdamW and Adafactor, while strictly bounding update volatility. The frameworkprovides a stable, low-latency surrogate possibly suited for large-scale decoupled architec-tures.

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